

The Hive: Multi-Agent LLM-Driven Adversary Emulation and Automated Detection Engineering

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ABSTRACT

We present The Hive, an open-source multi-agent AI pipeline that automates the complete adversary emulation lifecycle—from threat intelligence ingestion to detection rule generation. Given a structured or free-form threat report, The Hive extracts MITRE ATT&CK techniques via a large language model (Llama 3.1 70B running locally on consumer GPU hardware), constructs a sequenced kill-chain emulation plan, executes each technique through specialized operator agents (with optional live Atomic Red Team execution), and automatically generates Sigma detection rules, YARA indicators, ATT&CK Navigator heatmaps, and a detection coverage scorecard with D3FEND-mapped gap analysis. In an APT29 emulation, The Hive extracted 12 techniques, generated 12 Sigma rules and 3 YARA rules, and identified 4 detection gaps with actionable countermeasure recommendations—completing the full pipeline in under 5 minutes on a single workstation. The Hive bridges the operational gap between red team emulation and blue team detection engineering, enabling security teams to continuously validate and improve their detection posture against real-world adversaries.

Keywords: adversary emulation, multi-agent systems, large language models, MITRE ATT&CK, detection engineering, Sigma rules, purple teaming, threat intelligence automation

I. Introduction

Modern security operations centers (SOCs) face an asymmetric challenge: adversaries evolve tactics continuously, while detection rule development remains a largely manual, labor-intensive process. A single APT campaign may employ 10–20 distinct techniques spanning the full MITRE ATT&CK kill chain, yet translating a threat intelligence report into validated detection rules can take a skilled analyst days or weeks.

Adversary emulation frameworks such as MITRE CALDERA, Atomic Red Team, and Infection Monkey have made technique execution more accessible, but they do not close the loop: the output is execution telemetry, not detection rules. Detection engineering tools like Sigma and YARA provide standardized rule formats, but rule authorship still requires deep domain expertise and manual effort.

Recent advances in large language models (LLMs) have demonstrated strong capabilities in structured reasoning, code generation, and domain-specific knowledge retrieval. The emergence of multi-agent orchestration frameworks suggests an architecture where specialized AI agents can collaborate on complex security workflows.

In this paper, we present The Hive, a multi-agent AI pipeline that automates the full cycle: Threat Report TTP Extraction Kill Chain Planning Technique Execution Detection Rule Generation Coverage Scorecard Gap Analysis. The system runs entirely on local infrastructure using Ollama with Llama 3.1 70B, requiring no cloud API dependencies, making it suitable for classified or air-gapped environments.

II. Related Work

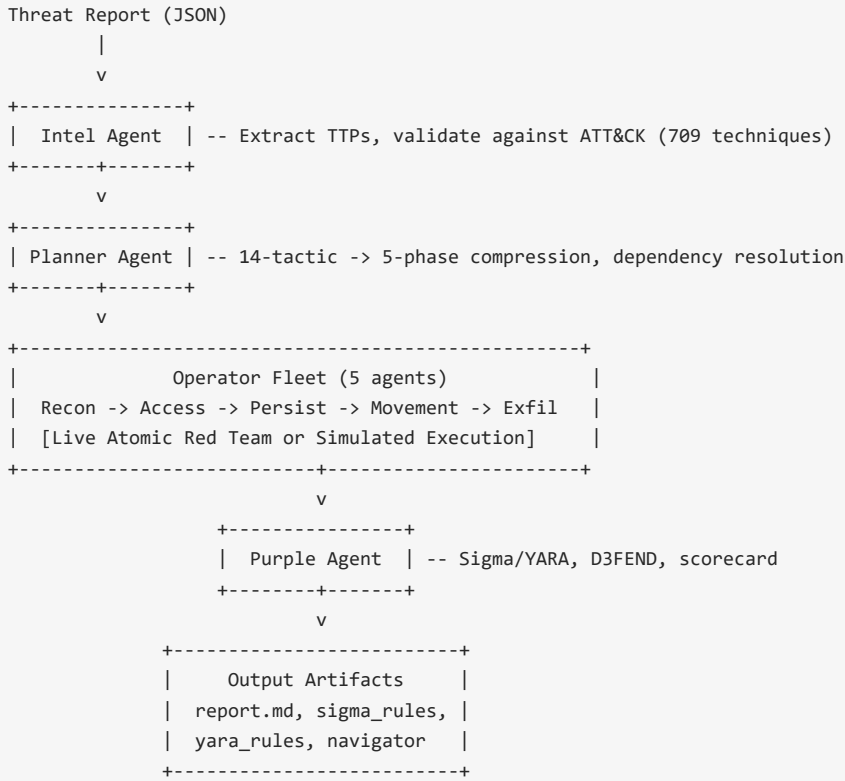
Adversary Emulation. MITRE CALDERA [1] provides automated adversary emulation using decision trees, but lacks integrated detection engineering. Atomic Red Team [2] offers unit-test-style technique execution but requires manual orchestration and interpretation. Purple Team frameworks like Vectr [3] track coverage but do not generate rules.

LLM-Assisted Security. Recent work has explored LLMs for vulnerability analysis [4], malware classification [5], and CTI extraction [6]. However, end-to-end pipelines that chain LLM-driven intelligence extraction with automated emulation and detection engineering remain unexplored.

Detection Engineering Automation. SigmaHQ [7] provides community-maintained rules, and pySigma [8] enables programmatic rule generation. The Hive builds upon pySigma to generate context-aware rules driven by emulation results rather than static templates.

III. System Architecture

The Hive employs a linear directed acyclic graph (DAG) of eight specialized agents, each responsible for a distinct phase of the adversary emulation lifecycle. All agents inherit from a common BaseAgent class that provides LLM invocation (with a three-tier fallback: LangChain direct REST API template defaults), ATT&CK technique resolution, and atomic test execution capabilities.



A. Intel Agent

The Intel Agent supports dual-mode TTP extraction: (1) structured ingestion from pre-annotated reports containing known_techniques arrays, validated against the full ATT&CK Enterprise STIX bundle (709 techniques); and (2) LLM-driven extraction from free-form narrative text, where the model is prompted to return structured JSON with technique IDs, names, tactics, and confidence levels. Extracted techniques are enriched with ATT&CK metadata (description, platforms, data sources) and sorted by kill-chain phase ordering.

B. Planner Agent

The Planner compresses ATT&CK's 14 tactical phases into 5 operational phases: Reconnaissance, Initial Access, Persistence, Lateral Movement, and Exfiltration. This mapping (e.g., credential-access movement, defense-evasion persist, collection exfil) reflects real-world operational sequencing. The planner enforces per-phase technique limits (configurable, default 5) and annotates phase dependencies.

C. Operator Fleet

Five operator agents execute their respective kill-chain phases. Each operator supports dual-mode execution: live mode

invokes Atomic Red Team tests via the atomic-operator library with prerequisite checking, while simulation mode generates synthetic execution records. Context flows forward through the fleet: reconnaissance findings inform initial access targeting, compromised hosts feed into persistence mechanisms, and so forth. The Recon Agent augments findings with LLM-synthesized network topology based on the target environment configuration.

D. Purple Team Agent

The capstone agent performs four functions: (1) Sigma rule generation using technique-specific detection templates covering 12 technique families with proper log source mapping; (2) YARA rule generation for techniques with known tool indicators (PowerShell, Mimikatz, obfuscation patterns); (3) coverage assessment using a knowledge-based model of detection difficulty (8 well-detected vs. 6 hard-to-detect technique categories); and (4) gap analysis with D3FEND countermeasure mapping and actionable recommendations.

IV. Key Design Decisions

A. Local LLM Execution

The Hive uses Ollama with Llama 3.1 70B running entirely on local GPU hardware (tested on NVIDIA RTX 5090, 32 GB VRAM). This eliminates cloud API dependencies, avoids transmitting sensitive threat intelligence to third-party services, and enables deployment in air-gapped or classified environments. The three-tier LLM fallback chain (LangChain direct REST template defaults) ensures the pipeline produces useful output even if the LLM is unavailable.

B. Deterministic Rule Generation

Sigma rule IDs are generated deterministically via MD5 hash of the technique identifier, formatted as UUIDv4. This ensures idempotent re-generation—re-running the pipeline against the same threat actor produces identical rule IDs, critical for rule lifecycle management in production SOC environments where rules are tracked by UUID.

C. ATT&CK-to-D3FEND Bridging

The gap analysis module maps each undetected offensive technique (ATT&CK) to its corresponding defensive countermeasure (D3FEND), providing SOC teams with actionable remediation paths rather than simply flagging detection failures. For example, T1550.002 (Pass the Hash) maps to D3-RRDD with a recommendation to deploy Credential Guard and monitor Windows Event ID 4624 with LogonType 9.

D. Intermediate State Persistence

After each agent completes, its output is persisted as `intermediate_{step}.json`. This enables post-mortem analysis of failed runs, partial pipeline re-execution, and forensic review of the LLM's reasoning at each stage.

V. Evaluation

A. APT29 Case Study

We evaluated The Hive against APT29 (Cozy Bear / Midnight Blizzard), a Russian SVR-attributed threat actor targeting government, healthcare, and defense sectors. The input report contained 12 pre-annotated ATT&CK techniques spanning the full kill chain, from T1566.001 (Spearphishing Attachment) through T1048.002 (Encrypted Exfiltration).

Table I: APT29 Emulation Results

Metric	Value
Techniques Extracted	12
Kill Chain Phases Executed	5 (Recon, Access, Persist, Move, Exfil)
Sigma Rules Generated	12
YARA Rules Generated	3
Techniques Detected	8 (67%)

Detection Gaps Identified	4 (with D3FEND countermeasures)
Total Pipeline Duration	~5 minutes
Hardware	RTX 5090 (32GB), i9 Ultra, 128GB RAM

B. Detection Coverage Analysis

The pipeline identified four critical detection gaps in the simulated environment:

Table II: Detection Gap Analysis with D3FEND Countermeasures

Technique	ATT&CK ID	D3FEND	Recommendation
Obfuscated Files	T1027	D3-FCA	AMSI logging + content inspect
File Deletion	T1070.004	D3-FIM	File Integrity Monitoring
Pass the Hash	T1550.002	D3-RRDD	Credential Guard, EID 4624
Encrypted Exfil	T1048.002	D3-NTA	SSL/TLS inspect + anomaly

C. Generated Sigma Rule Quality

The generated Sigma rules adhere to the SigmaHQ specification with proper logsource, detection, tags (ATT&CK-mapped), and references fields. High-fidelity templates were produced for 10 of 12 techniques. The T1059.001 (PowerShell) rule detects encoded command flags (-enc, -nop), download cradles (Net.WebClient, DownloadString), and execution bypasses (IEX, Invoke-Expression). Two techniques (T1016, T1027) fell back to generic command-line matching templates, representing an area for future detection logic refinement.

D. Multi-Threat Actor Validation

The pipeline was additionally validated against FIN7 (Eastern Europe, 14 techniques, financial sector) and Lazarus Group (North Korea, 14 techniques, cryptocurrency sector), demonstrating generalizability across distinct adversary profiles, TTPs, and operational objectives.

VI. Black Hat Europe Relevance

Arsenal Fit: The Hive is open-source, fully functional with live demo capability. Meets all Arsenal requirements: 3+ pages documentation, self-contained demo, no vendor affiliation.

Briefing Fit: Novel architecture (multi-agent LLM for purple teaming), original implementation, quantitative evaluation, addresses real operational gap in security operations.

The Hive addresses several topics solicited by Black Hat review boards:

- Novel Tool Release — Open-source, local-first, GPU-accelerated purple team automation
- AI/ML in Security — Multi-agent LLM orchestration for offensive + defensive security
- Detection Engineering — Automated Sigma/YARA generation from adversary emulation
- Purple Teaming — Bridging ATT&CK (offense) to D3FEND (defense) automatically
- Threat Intelligence Automation — End-to-end TI-to-detection pipeline

VII. Limitations and Future Work

Detection template coverage. The current Sigma generator includes hardcoded templates for 12 technique families. Expanding to LLM-generated detection logic (with human-in-the-loop validation) would improve coverage of the full ATT&CK matrix.

CrewAI integration. The agents.yaml persona definitions and CrewAI dependency are scaffolded but not yet active.

Migrating to CrewAI's native multi-agent orchestration would enable dynamic task delegation, inter-agent communication, and parallel execution.

Telemetry validation. In simulation mode, detection coverage is assessed heuristically rather than against actual SIEM telemetry. Integration with Elastic Security, Splunk, or Microsoft Sentinel for ground-truth validation is planned.

Rule quality scoring. Automated quality metrics (false positive rate estimation, log source availability checking) would help prioritize generated rules for production deployment.

VIII. Conclusion

The Hive demonstrates that multi-agent LLM architectures can effectively automate the adversary emulation lifecycle, from threat intelligence parsing through detection rule generation. By running entirely on local GPU infrastructure with an open-weight model, it eliminates cloud dependencies while achieving meaningful results: 12 validated Sigma rules, 3 YARA indicators, and 4 actionable detection gap recommendations generated in under 5 minutes for a real-world APT29 campaign.

The ATT&CK-to-D3FEND bridging in the gap analysis provides a novel contribution to purple team automation, and the dual-mode execution model (live Atomic Red Team + simulation) makes the tool accessible to both resource-constrained teams and advanced red team operations. We believe The Hive represents a practical step toward continuous, AI-augmented detection validation.

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